Computer Software



Objectives Overview

- Explain software and identify the types of software
- Describe the functions of an operating system
- Summarize the startup process on a personal computer
- Summarize the features of stand-alone operating systems
- Briefly describe various server operating systems
- Summarize the features of embedded operating systems
- Explain the purpose of utility programs

Software

Software is a collection of instructions that tell a computer how to perform specific tasks or functions.

- Responsible for enabling computers to perform a wide range of operations:
  - running of applications (executing algorithms)
  - managing hardware devices
  - processing data

There are two (2) main types of software: System and Application Software.

NB: Also two (2) other software: Middleware and Firmware

Software

System Software: includes operating systems like Windows, macOS, and Linux.

- It manages computer hardware resources and provides essential services to other software.
- Device drivers, which enable communication between the operating system and hardware devices, and utility programs are part of system software.

Application software: consists of programs designed for specific tasks or purposes.

- Examples include word processors (e.g., Microsoft Word), web browsers (e.g., Google Chrome), spreadsheet software (e.g., Microsoft Excel), and specialized applications for graphics, video editing, and gaming.

Software

Middleware : acts as a mediator between different software applications.

- It facilitates communication and data exchange between software components.
- Middleware is commonly used in networked and distributed computing environments to enable seamless interactions between software systems.

Firmware: a specialized type of software embedded in hardware devices.

- It can be found in components like computer BIOS (Basic Input/Output System), printers, and routers, etc.
- Firmware provides low-level instructions and control for hardware components, ensuring their proper functionality.

System Software

System software consists of the programs that control or maintain the operations of the computer and its devices.







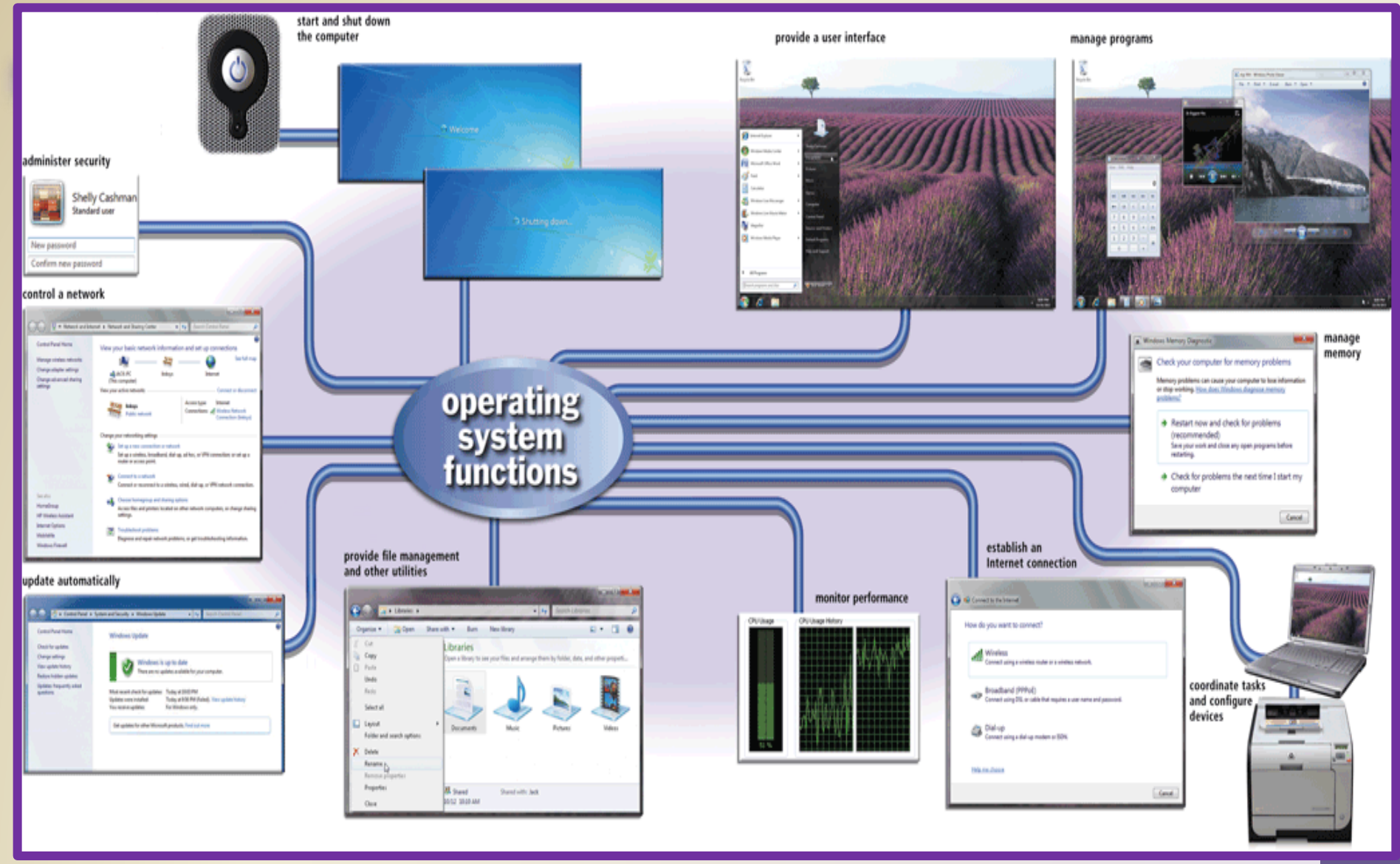
Operating Systems

- An **Operating System (OS)** is a set of programs containing instructions that work together to coordinate all the activities among computer hardware resources. *NB: The activities could be among computer or mobile device hardware.*

Functions of the OS

Start and shut down a computer	Provide a user interface	Manage programs	Manage memory
Coordinate tasks	Configure devices	Establish an Internet connection	Monitor performance
Provide utilities	Automatically update	Control a network	Administer security

Operating Systems



Operating System Functions

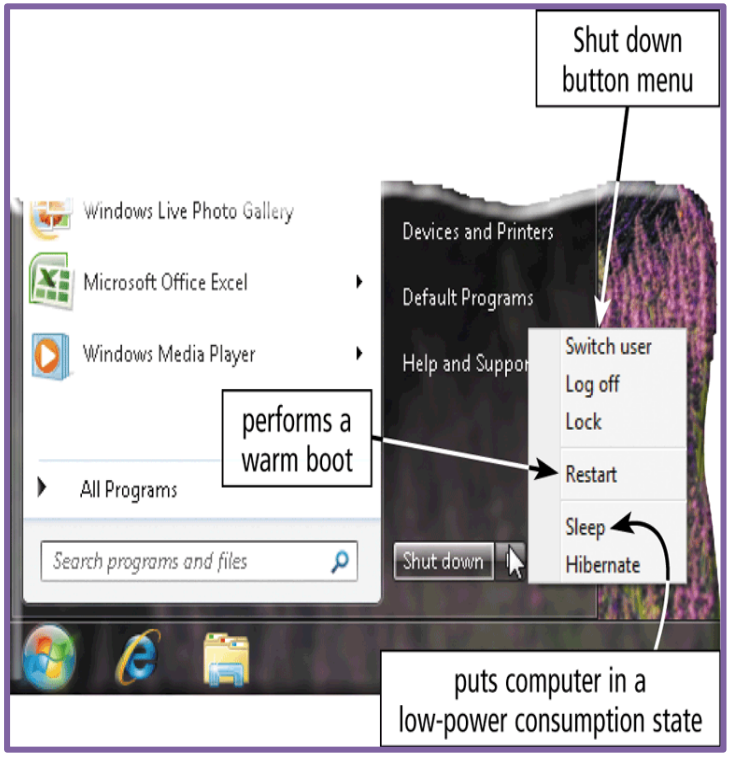
- The process of starting or restarting a computer is called **booting.**

Cold boot

- Turning on a computer that has been powered off completely.

Warm boot

- Using the operating system to restart a computer.



Operating System Functions

The Booting Process:

Step 1: When you turn on the computer or mobile device, the power supply or battery sends an electrical current to the circuits in the computer or mobile device.

Step 2: The charge of electrical current causes the processor chip to reset itself and finds the firmware that contains start-up instructions.

Step 3: The start-up process executes a series of tests to check the various components (POST).

POST – Power-On-Self-Test.

Operating System Functions

The Booting Process:

Step 4: If the tests are successful, **the kernel** of the operating system and other **frequently used instructions** are loaded from the computer or mobile device's internal storage media to its memory (RAM).

NB: The ***kernel* is the core** of an operating system that manages memory and devices, maintains the internal clock, runs programs, and assigns the resources, such as devices, programs, apps, data, and information.

Operating System Functions

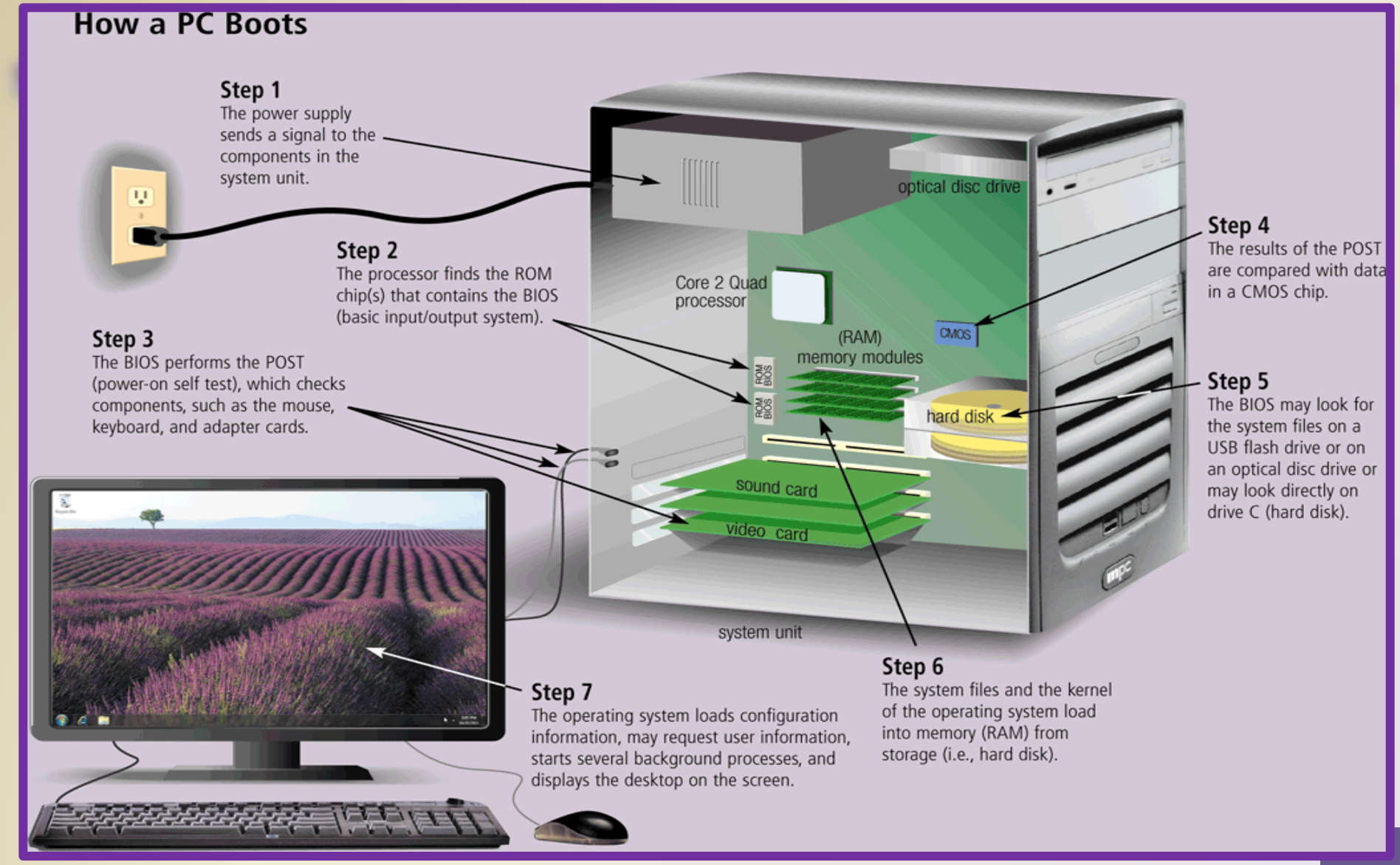
The Booting Process:

Step 5: The operating system in memory **takes control** of the computer or mobile device and loads system configuration information.

The operating system may **verify** that the person attempting to use the computer or mobile device is a legitimate user.

Finally, the **user interface appears on the screen**, and any start-up applications, such as antivirus software, run.

Operating System Functions



Operating System Functions

A **boot drive** is the drive from which your computer starts.

- You can boot from a **boot disk.**
- A **recovery disk** contains a few system files that will start the computer.

Operating System Functions

- An operating system includes various shut down options.

Sleep mode saves any open documents and programs to **RAM**, turns off all unneeded functions, and then places the computer in a low-power state

Hibernate saves any open documents and programs to a **hard disk** before removing power from the computer

Operating System Functions

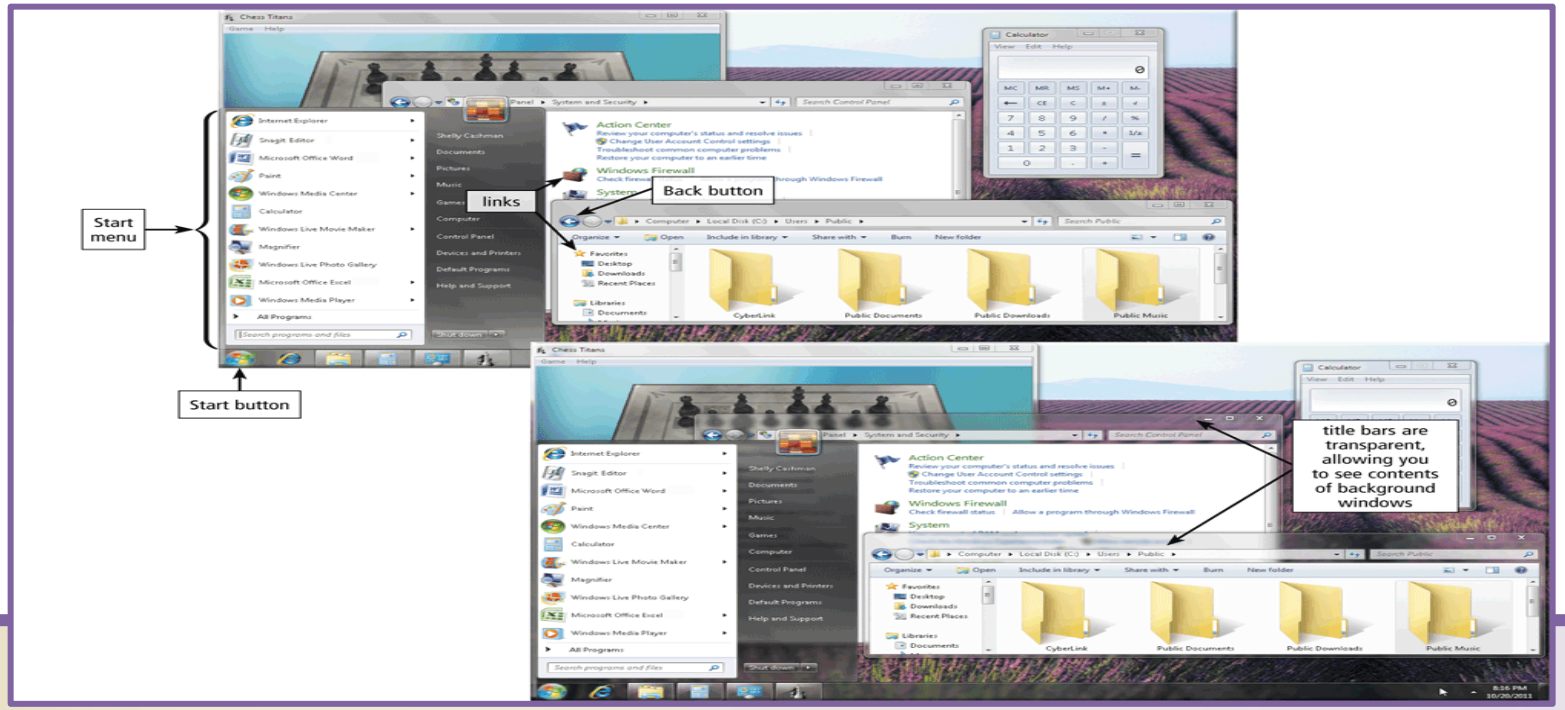
A **user interface** controls how you enter data and instructions and how information is displayed on the screen.

- **Command-line**
- **Graphical User Interface** (GUI) and
- **Natural User Interface** (NUI)

Operating System Functions

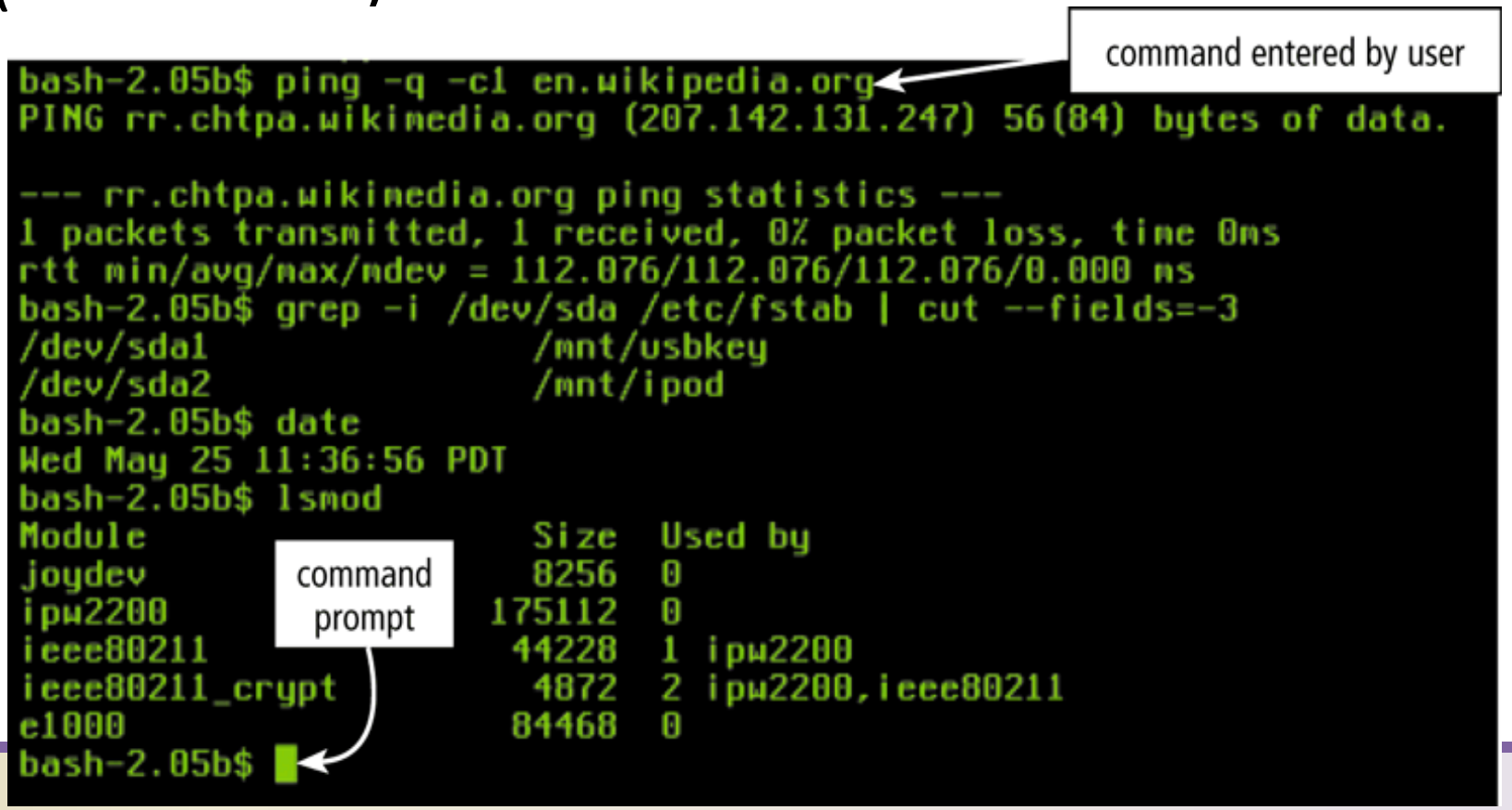
With a **Graphical User Interface (GUI):** the **user** interact with menus and visual images (icons).

The keyboard, mouse and other input devices could be used for interaction.



Operating System Functions

- With a **command-line** interface, a user uses the keyboard to enter data and instructions (*commands*).



Operating System Functions

Natural user interface (NUI), users interact with the software through ordinary, intuitive behaviour.

NUIs are implemented in a variety of ways: **touch screens** (touch input), **gesture recognition** (motion input), **speech recognition** (voice input), and **virtual reality** (simulations).

Operating System Functions

- How an operating system handles programs directly affects the productivity of the user.

Single user and multiuser

Single tasking and multitasking

Foreground and background

Preemptive multitasking

Multiprocessing

Single user and multiuser

Single user OS: supports only one user at a time.

Multiuser operating system enables two or more users to run programs simultaneously.

- Networks, servers, and supercomputers allow hundreds to thousands of users to connect at the same time and, thus, use multiuser operating system.

Single tasking and multitasking

Single tasking operating system: allows only one program or app to run at a time.

Operating systems on **embedded computers** and **some mobile devices** use a single tasking operating system.

Multitasking operating system: allows two or more programs or apps to reside in memory at the same time. Both programs can run concurrently.

Foreground and background

Foreground program is the active program, which is currently being used by the user of the computer.

The foreground program typically is displayed on the screen.

Background programs are programs that are running but are not displayed on the screen.

Background programs are hidden partially or completely behind the foreground program.

Single tasking and multitasking

Preemptive multitasking: OS uses some criteria to decide on how long allocation of computer resources is done to **any one task** before giving **another task** its turn.

Preempting is the act of taking control of the operating system from one task and giving it to another task.

Multiprocessing

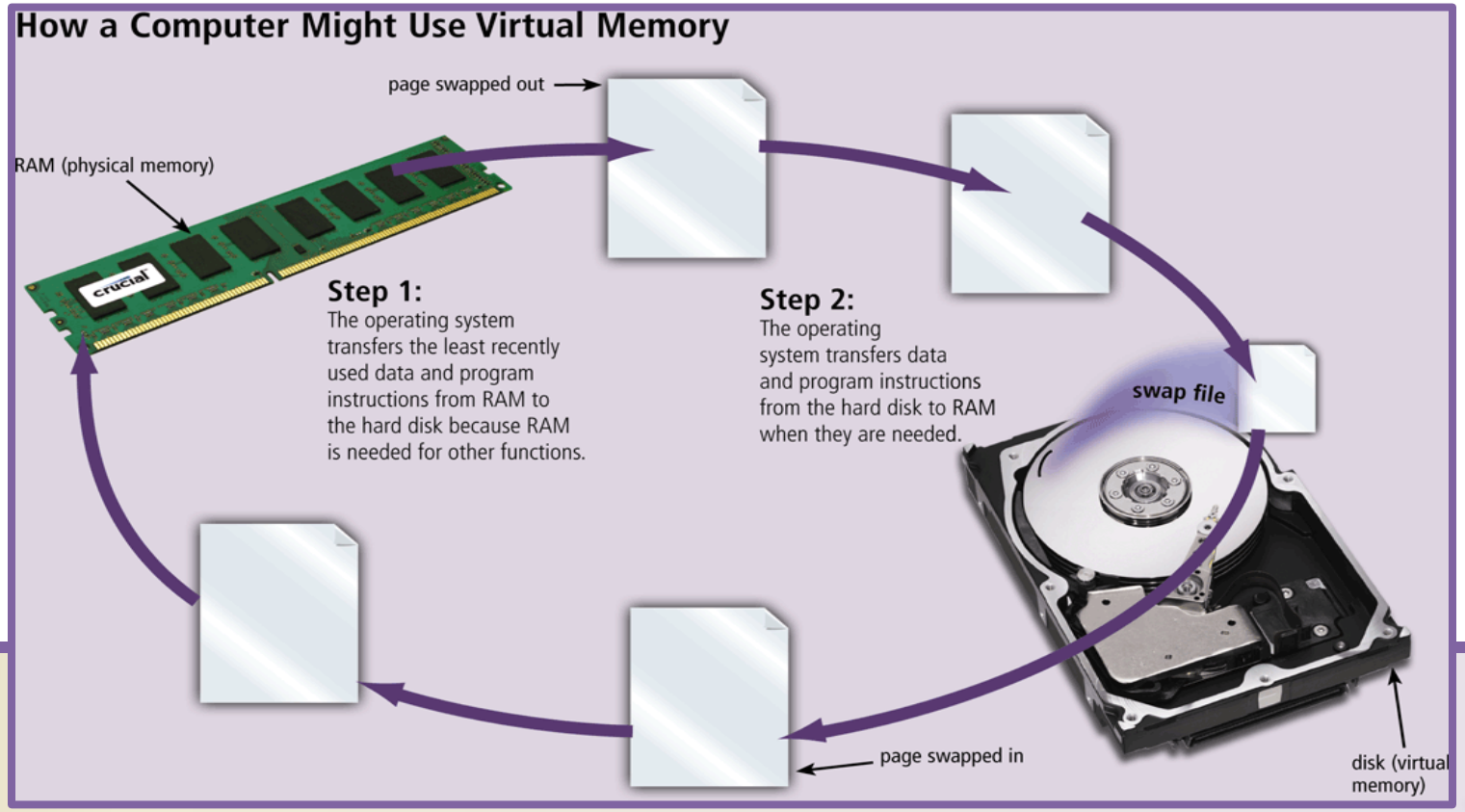
A **multiprocessing operating system** supports **two or more** processors running programs at the same time.

Multiprocessing involves the **coordinated processing** of programs by more than one processor.

Multiprocessing increases a computer's **processing speed.**

Operating System Functions

- **Memory management** optimizes the use of **RAM**.
- **Virtual memory** is a portion of a storage medium functioning as additional **RAM**.



Operating System Functions

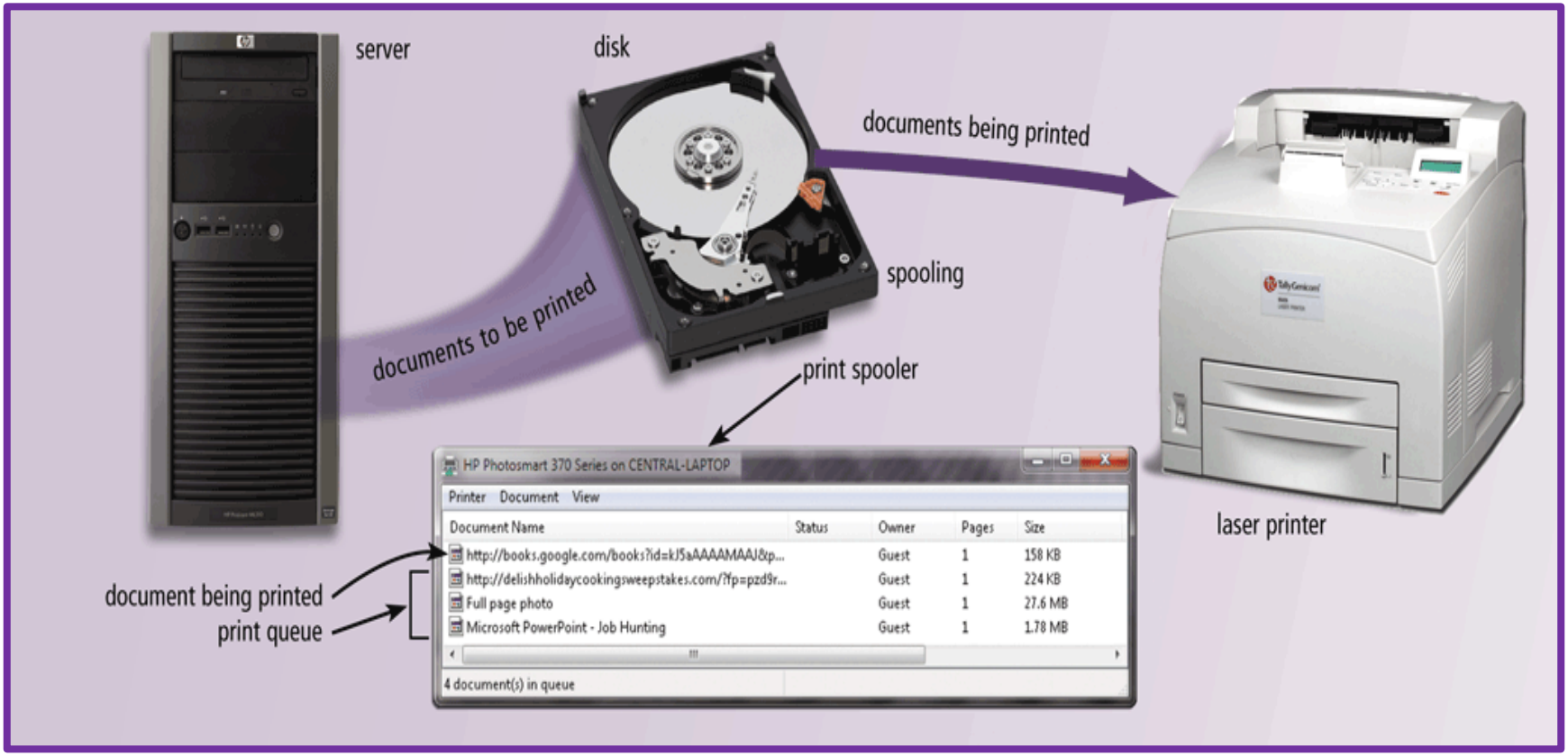
The area of the hard drive used for virtual memory is called a ***swap file***.

The technique of swapping items between memory and storage, called ***paging***.

When an operating system spends much of its time paging, instead of executing application software, it is said to be ***thrashing***.

Operating System Functions

- The operating system determines the order in which tasks are processed.



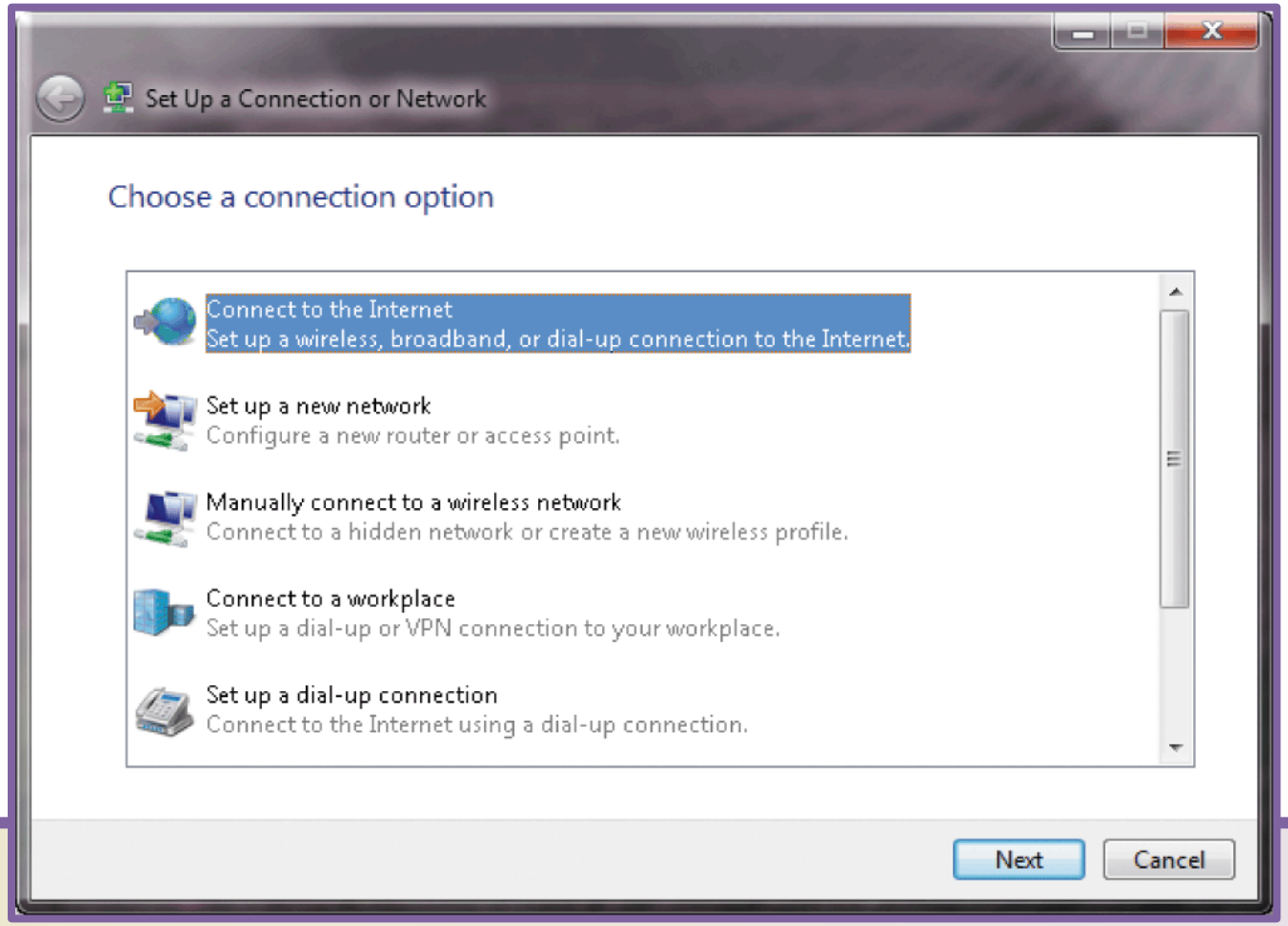
Operating System Functions

A **driver** is a small program that tells the operating system how to communicate with a specific device

Plug and Play automatically configures new devices as you install them

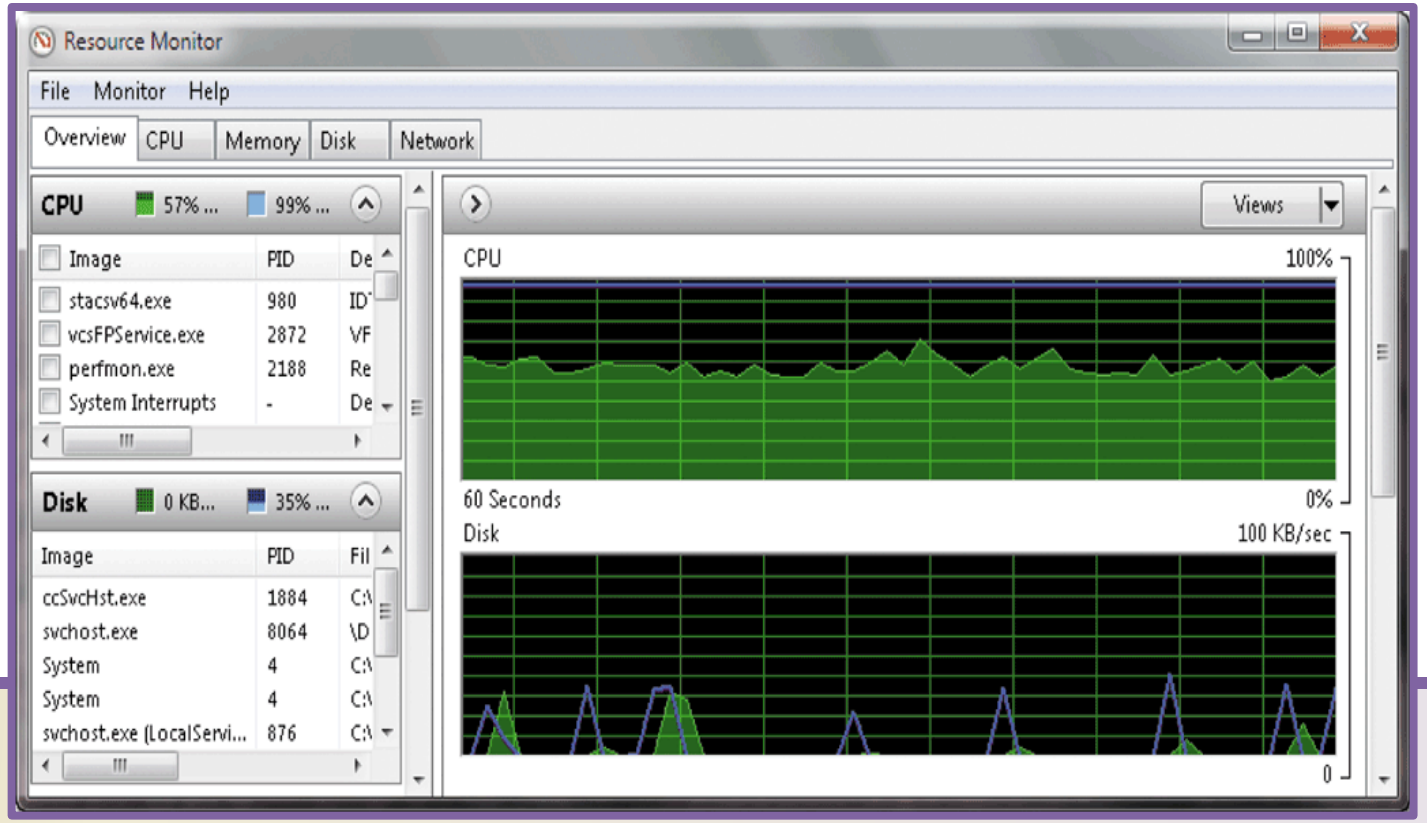
Operating System Functions

- Operating systems typically provide a means to **establish Internet connections**.



Operating System Functions

- A **performance monitor** is a program that assesses and reports information about various computer resources and devices.



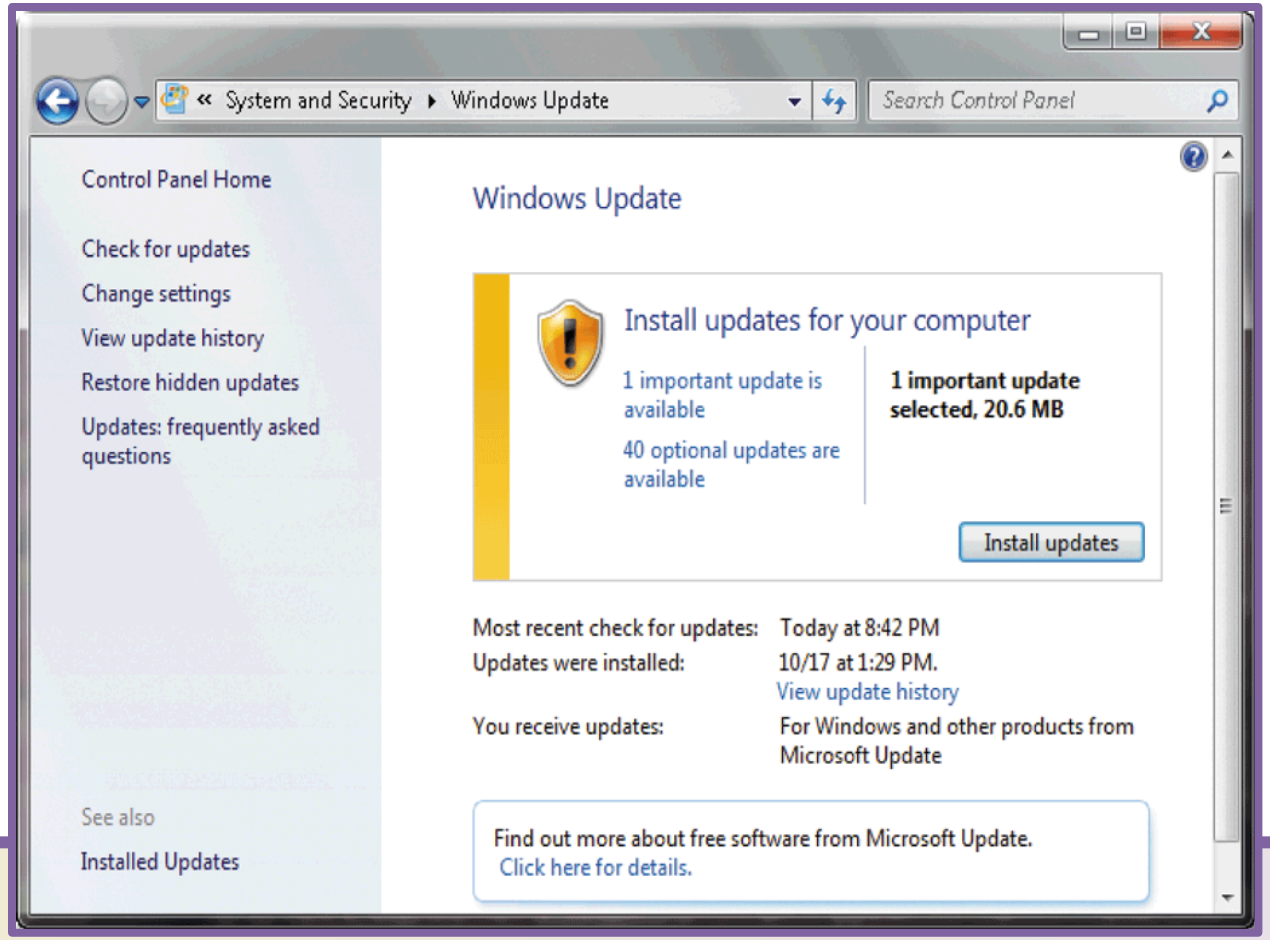
Operating System Functions

- Operating systems often provide users with the capability of:

Managing files	Searching for files	Viewing images	Securing a computer
Uninstalling programs	Cleaning up disks	Defragmenting disks	Diagnosing problems
	Backing up files and disks	Setting up screen savers	

Operating System Functions

- **Automatic update** automatically provides updates to the program.

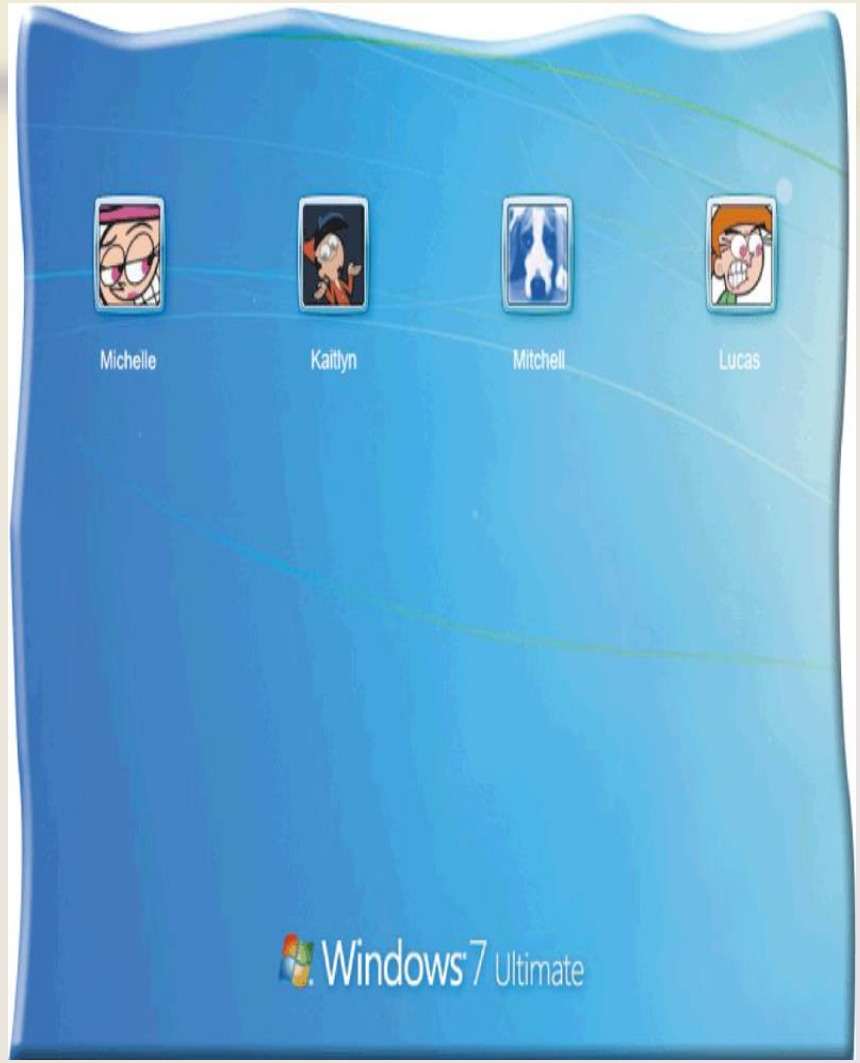


Operating System Functions

- **Server operating systems** are designed to work with a server on a network.
- A **server operating system** organizes and coordinates how multiple users access and share resources on a network.
- A **network administrator** uses the server operating system to:
  - Add and remove users, computers, and other devices
  - Install software and administer network security

Operating System Functions

- Each user has a user account:
  - A **user name**, or **user ID**, identifies a specific user
  - A **password** is a private combination of characters associated with the user name



Virtual machines

A **virtual machine** is a software-based emulation of a computer (hardware), allowing multiple operating systems or applications to run on a single physical machine.

The machine operates as though there are two or more separate and independent computers working.

The operating system of the physical machine is known as the **host operating system**.

The operating system for each virtual machine is known as the **guest operating system.**

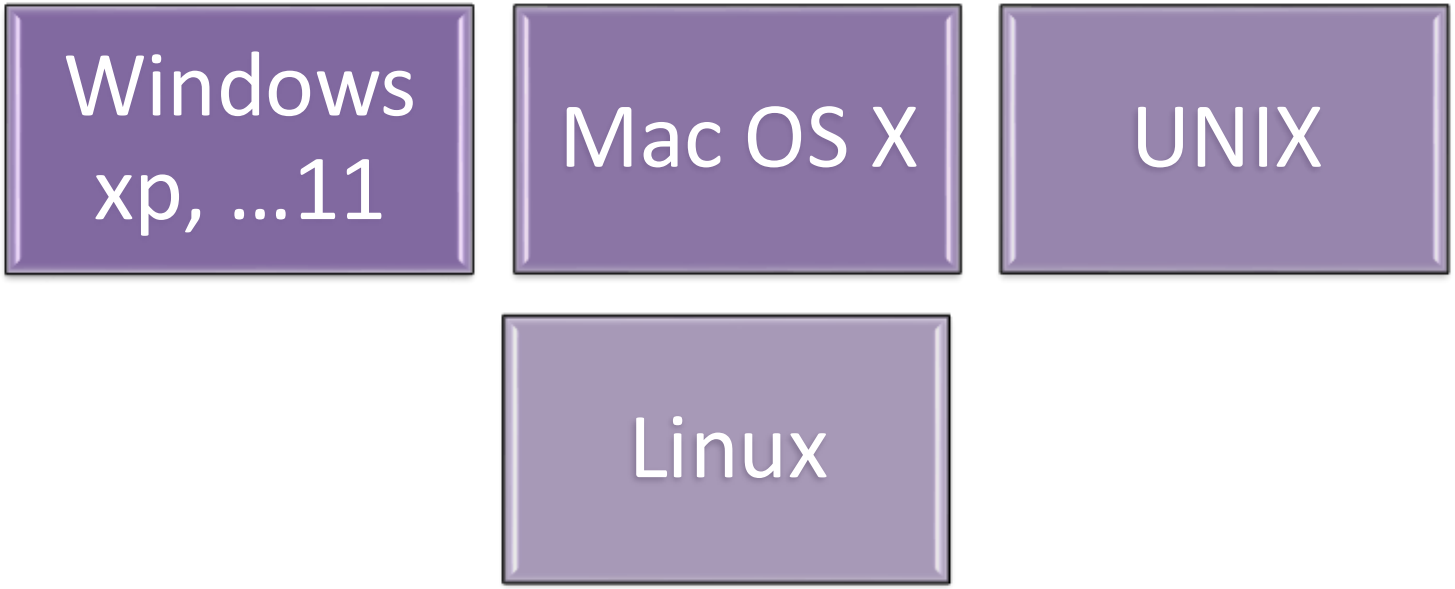
Types of Operating Systems

Categories of Operating Systems

Category	Operating System Name
Stand-alone	• DOS • Early Windows versions (Windows 3.x, Windows 95, Windows NT Workstation, Windows 98, Windows 2000 Professional, Windows Millennium Edition, Windows XP, Windows Vista) • Windows 7 • Mac OS X • UNIX • Linux
Server	• Early Windows Server versions (Windows NT Server, Windows 2000 Server, Windows Server 2003) • Windows Server 2008 • UNIX • Linux • Solaris • NetWare
Embedded	• Windows Embedded CE • Windows Phone 7 • Palm OS • iPhone OS • BlackBerry • Google Android • Embedded Linux • Symbian OS

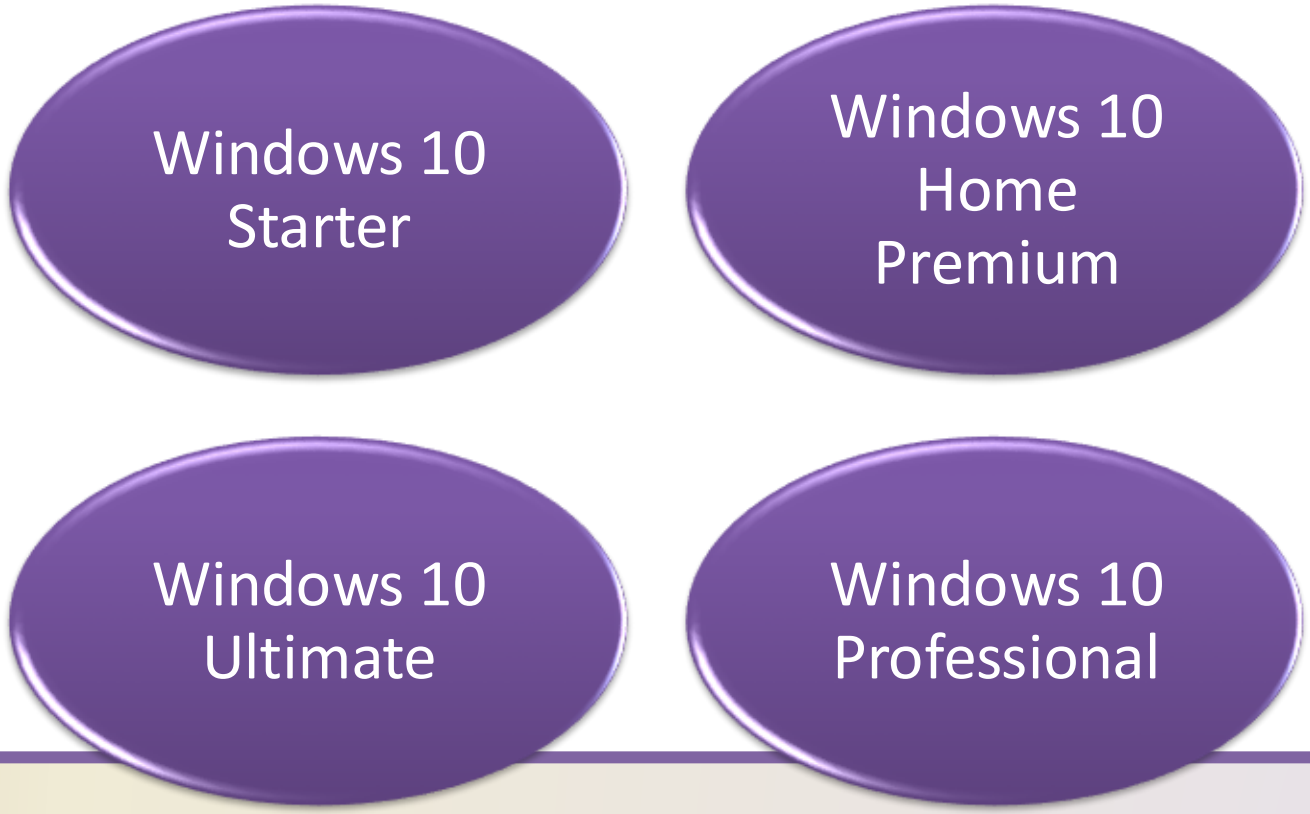
Stand-Alone Operating Systems

- A **stand-alone operating system** is a complete operating system that works on a desktop computer, notebook computer, or mobile computing device.



Stand-Alone Operating Systems

- **Windows 10** is Microsoft's fastest, most efficient operating system to date and is available in multiple editions:



Stand-Alone Operating Systems

- The **Macintosh operating system** has set the standard for operating system ease of use.
- Latest version is Mac OS X.



Stand-Alone Operating Systems



UNIX is a multitasking operating system developed in the early 1970s.



Linux is an open-source, popular, multitasking UNIX-type operating system.

Server Operating Systems

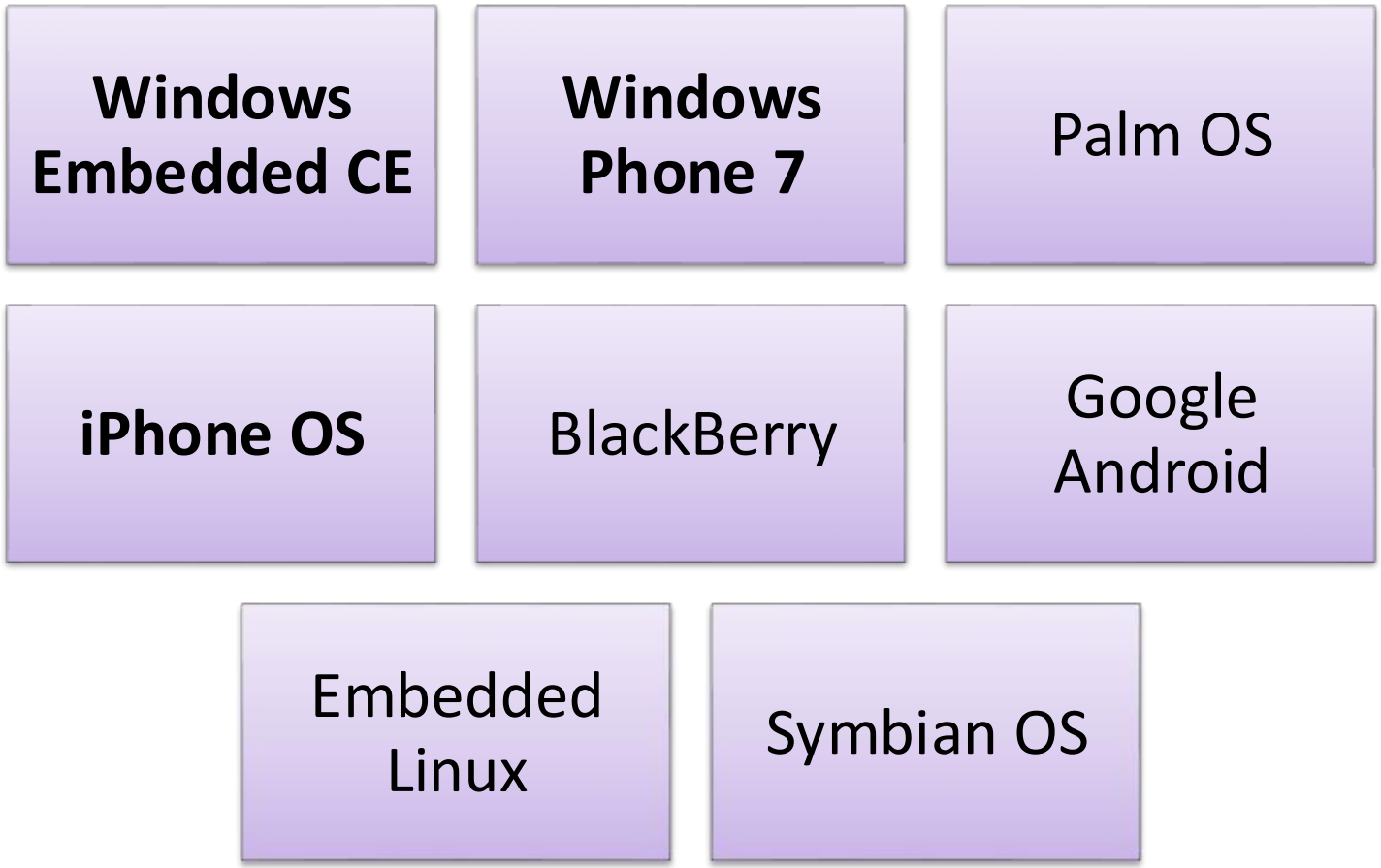
- Windows Server 2008
- UNIX
- Linux
- Solaris
- NetWare

Server Operating Systems

- **Windows Server 2008** is an upgrade to Windows Server 2003.
  - Part of the **Windows Server 2008 family**
  - Multiple editions
- Most editions include Hyper-V, a virtualization technology.
  - Virtualization is the practice of sharing or pooling computing resources

Embedded Operating Systems

- An **embedded operating system** resides on a ROM chip on a mobile device or consumer electronic device.



Embedded Operating Systems









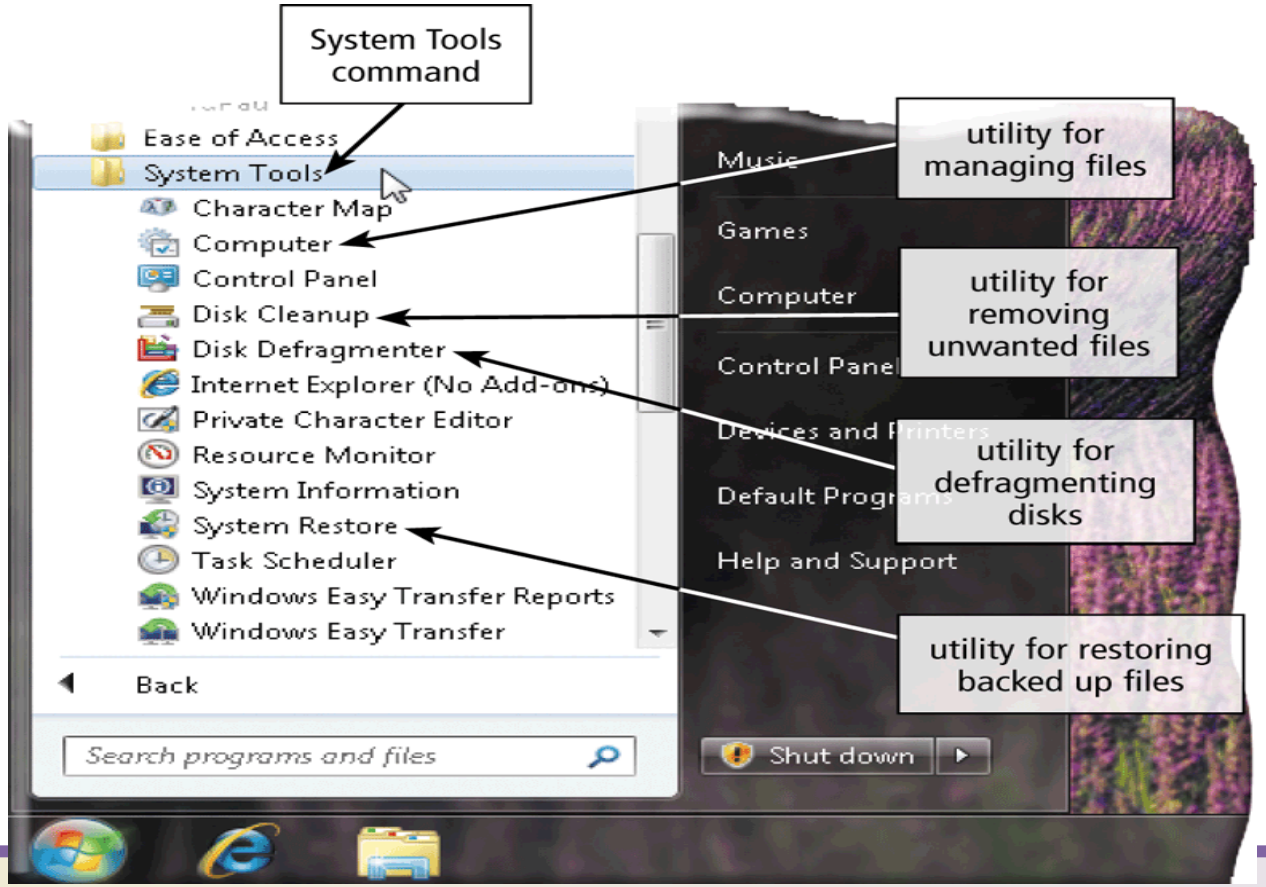






Utility Programs

- A **utility program** is a type of system software that allows a user to perform maintenance-type tasks.



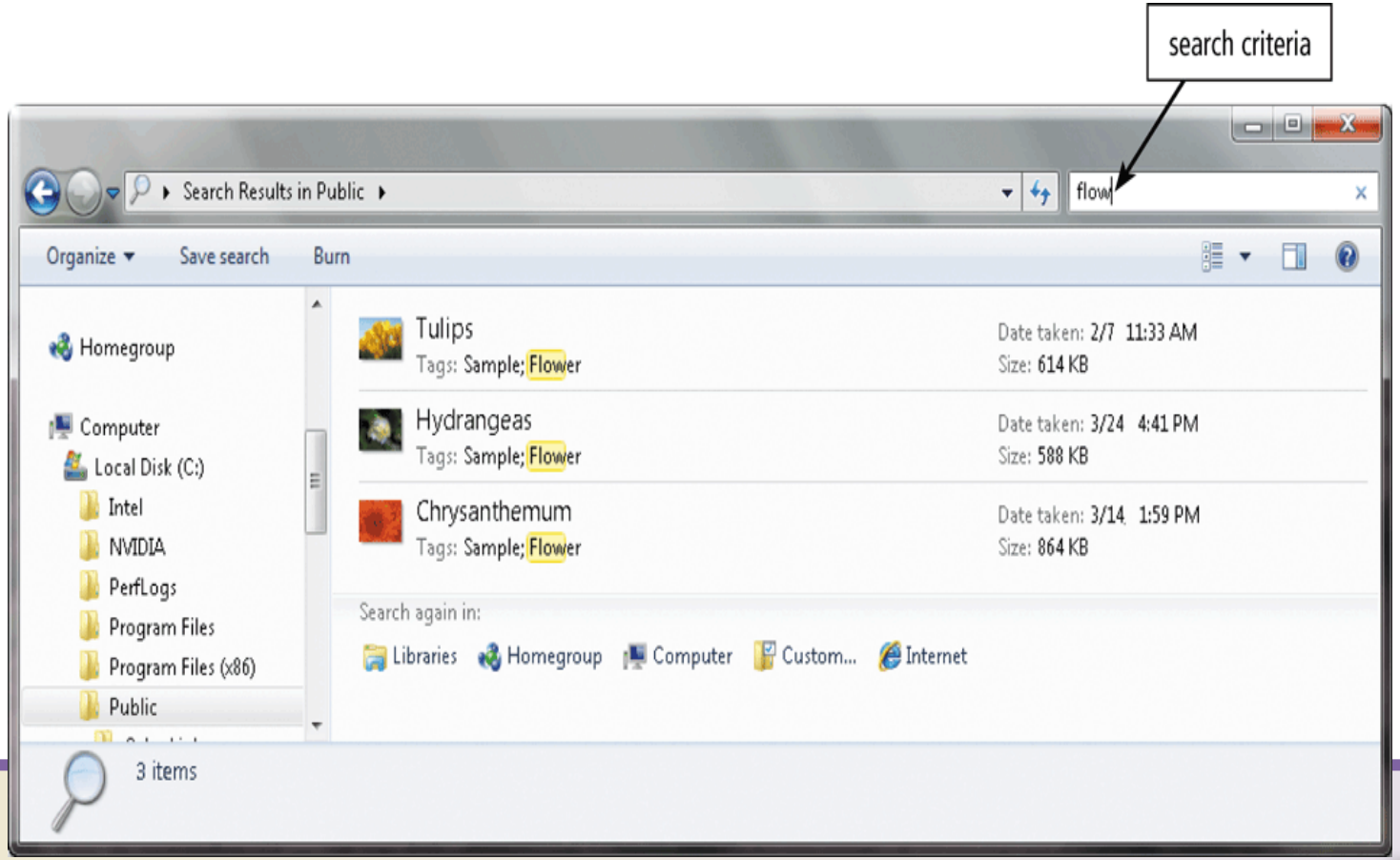
Utility Programs

- A **file manager** is a utility that performs functions related to file management:
  - Displaying a list of files
  - Organizing files in folders
  - Copying, renaming, deleting, moving, and sorting files and **folders**
  - Creating **shortcuts**



Utility Programs

- A **search utility** is a program that attempts to locate a file on your computer based on criteria you specify.



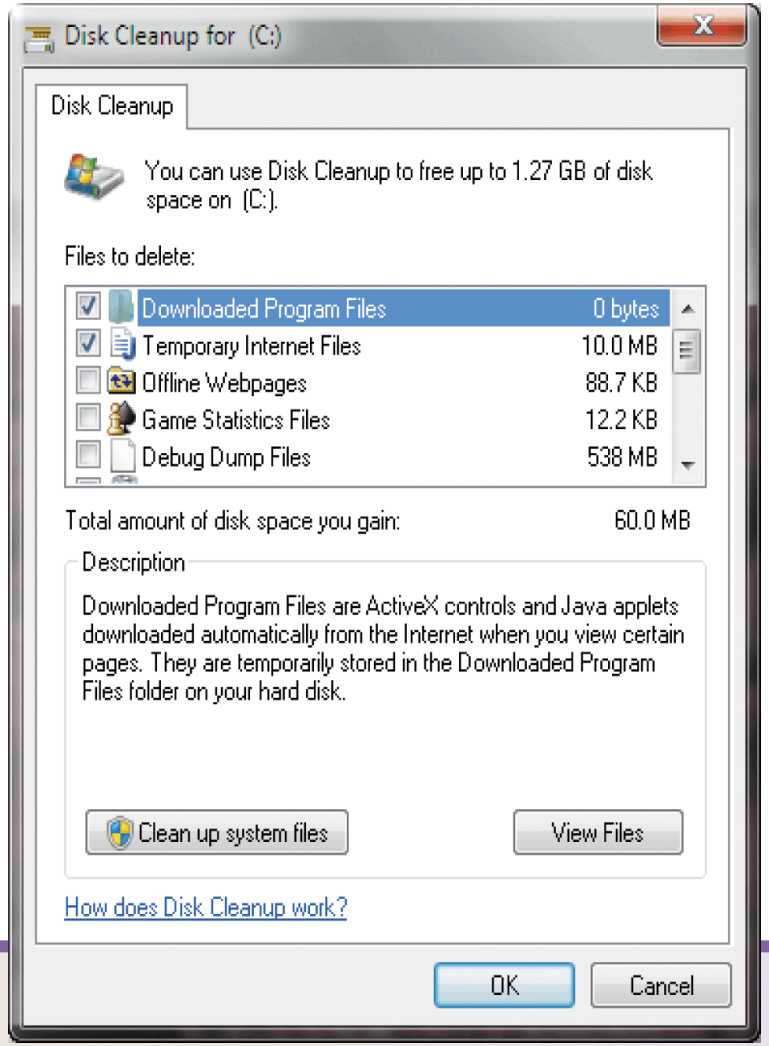
Utility Programs

- An **image viewer** allows users to display, copy, and print the contents of a graphics file.
- An **uninstaller** removes a program, as well as any associated entries in the system files.



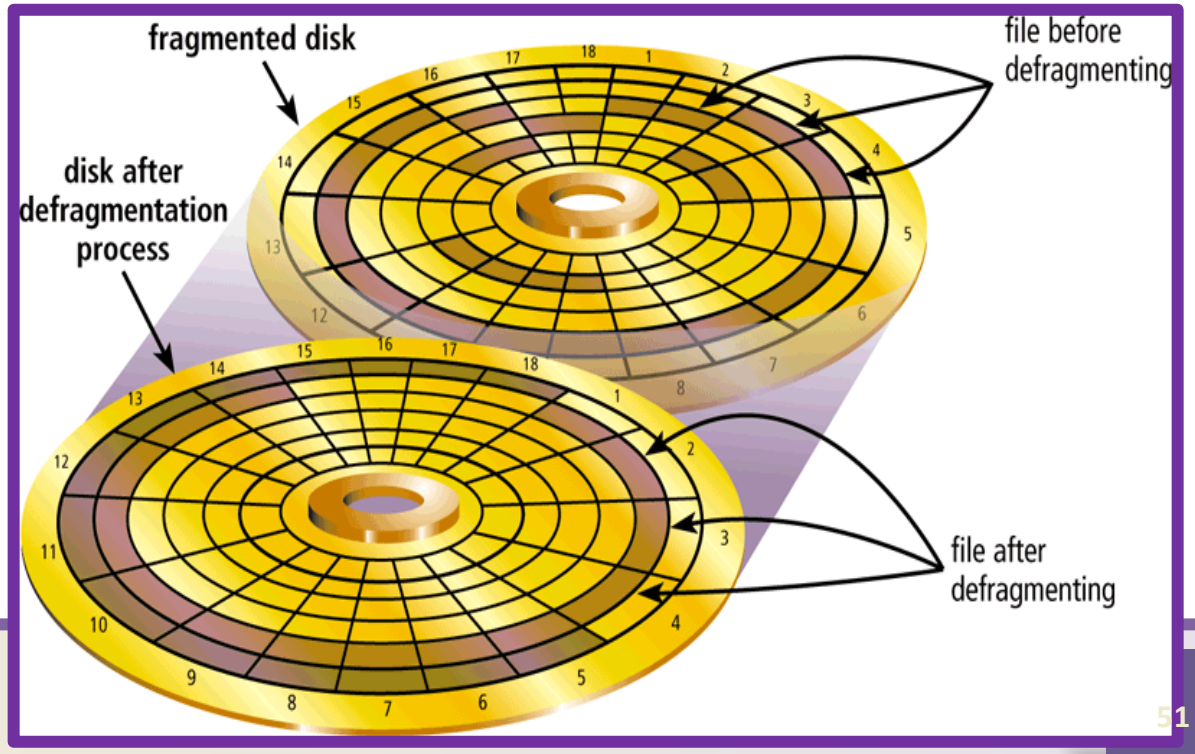
Utility Programs

- A **disk cleanup utility** searches for and removes unnecessary files:
  - Downloaded program files
  - Temporary Internet files
  - Deleted files
  - Unused program files



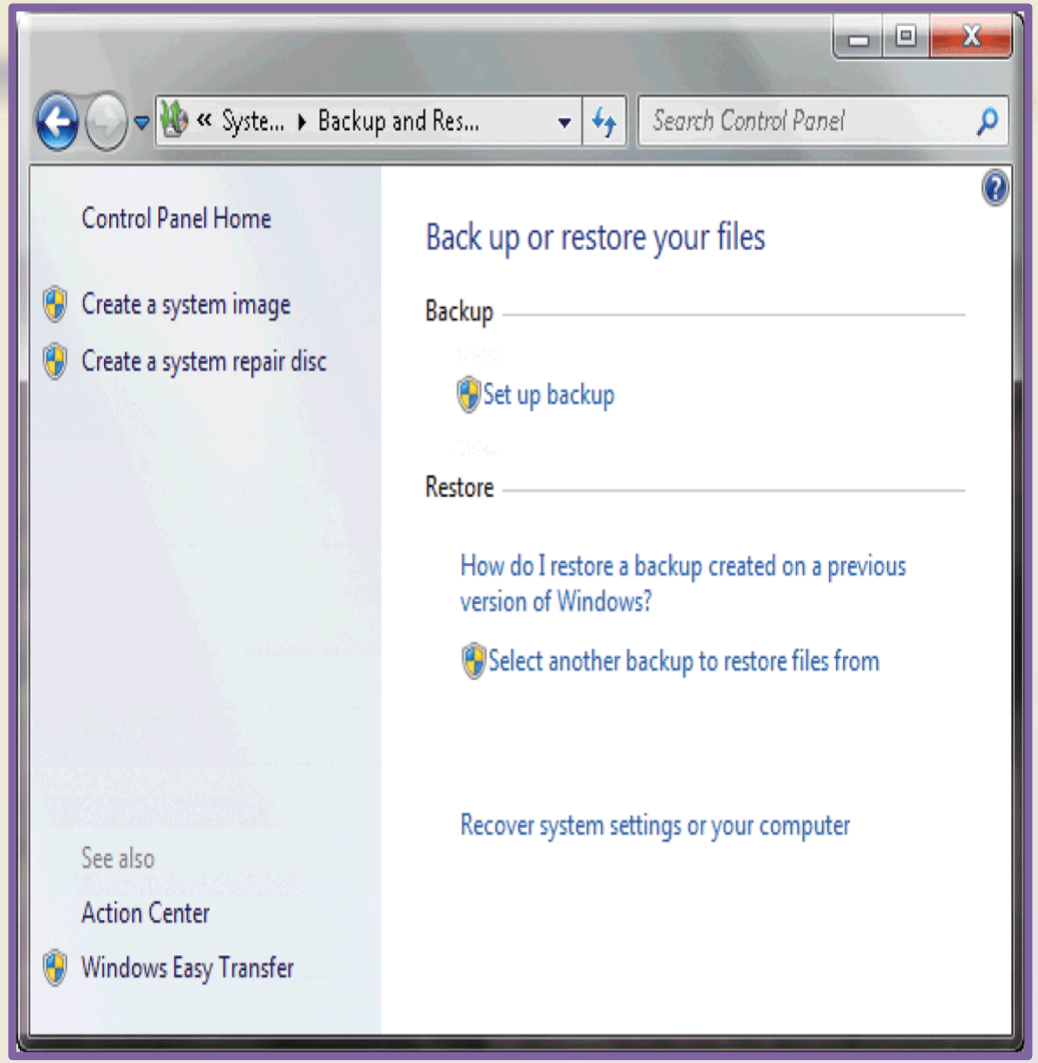
Utility Programs

- A **disk defragmenter** reorganizes the files and unused space on a computer's hard disk so that the operating system accesses data more quickly and programs run faster.
  - **Defragmenting**



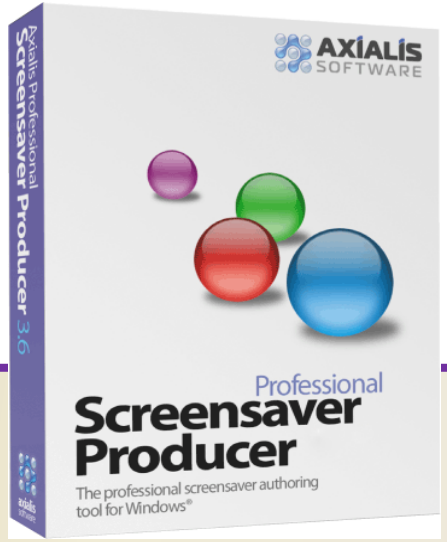
Utility Programs

- A **backup utility** allows users to copy files to another storage medium.
- A **restore utility** reverses the process and returns backed up files to their original form.

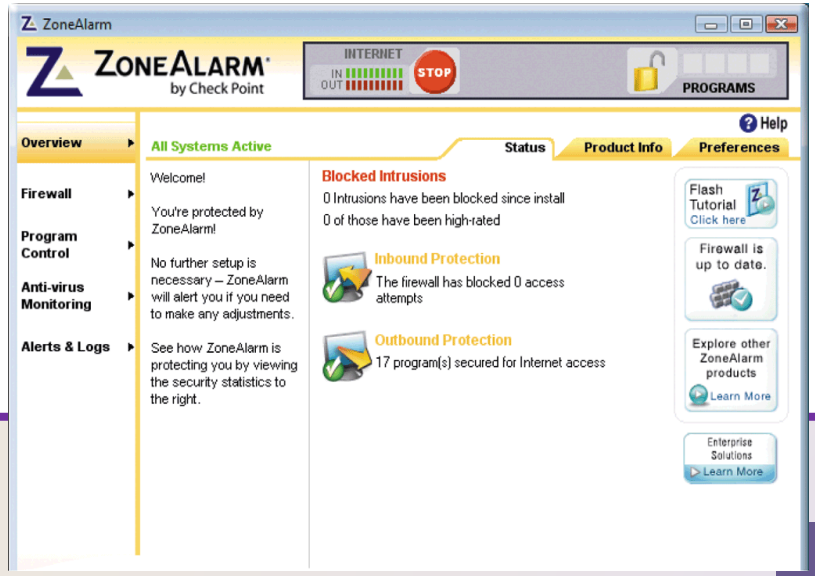


Utility Programs

- A **screen saver** causes a display device's screen to show a moving image or blank screen if no activity occurs for a specified time.



- A **personal firewall** detects and protects a personal computer from unauthorized intrusions.



Utility Programs

A **virus** describes a potentially damaging computer program that affects a computer negatively.

A **worm** copies itself repeatedly in memory or over a network.

A **Trojan horse** hides within or looks like a legitimate program.

An **antivirus program** protects a computer against viruses.

Utility Programs

Signs of Virus Infection

- An unusual message or image is displayed on the computer screen
- An unusual sound or music plays randomly
- The available memory is less than what should be available
- A program or file suddenly is missing
- An unknown program or file mysteriously appears
- The size of a file changes without explanation
- A file becomes corrupted
- A program or file does not work properly
- System properties change
- The computer operates much slower than usual



Utility Programs

Spyware is a program placed on a computer without the user's knowledge that secretly collects information about the user.

- A **spyware remover** detects and deletes spyware and other similar programs

Adware displays an online advertisement in a banner or pop-up window.

- An **adware remover** is a program that detects and deletes adware

Utility Programs

Filters are programs that remove or block certain items from being displayed:

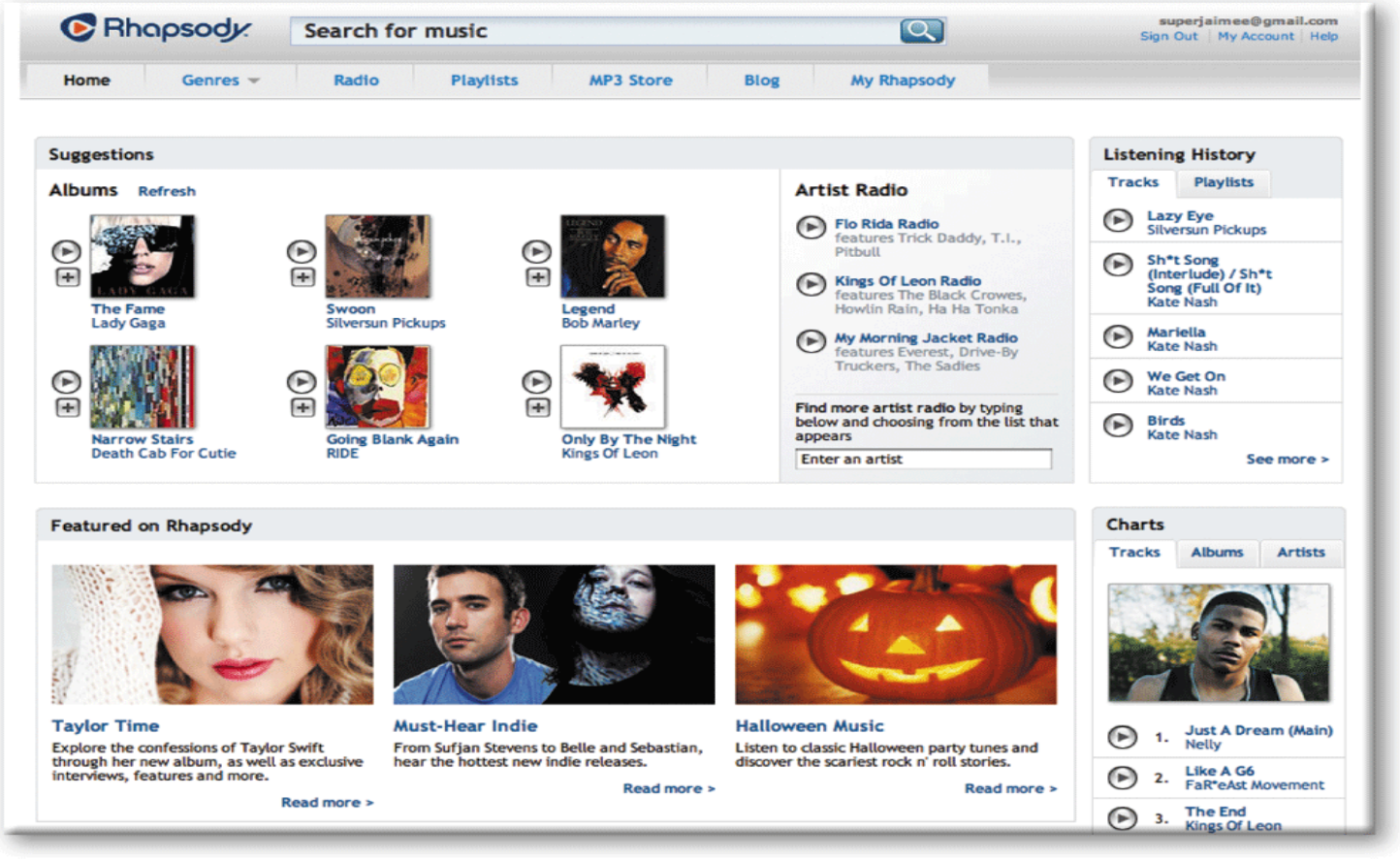
- **Web filtering software**
- **Anti-spam programs**
- **Phishing filters**
- **Pop-up blockers**

Utility Programs

- A **file compression utility** shrinks or reduces the size of a file(s):
  - Compressing files frees up space on the storage media
  - Two types of compression
    - Lossy
    - Lossless
- Compressed files sometimes are called **zipped files.**
  - Can be **uncompressed**

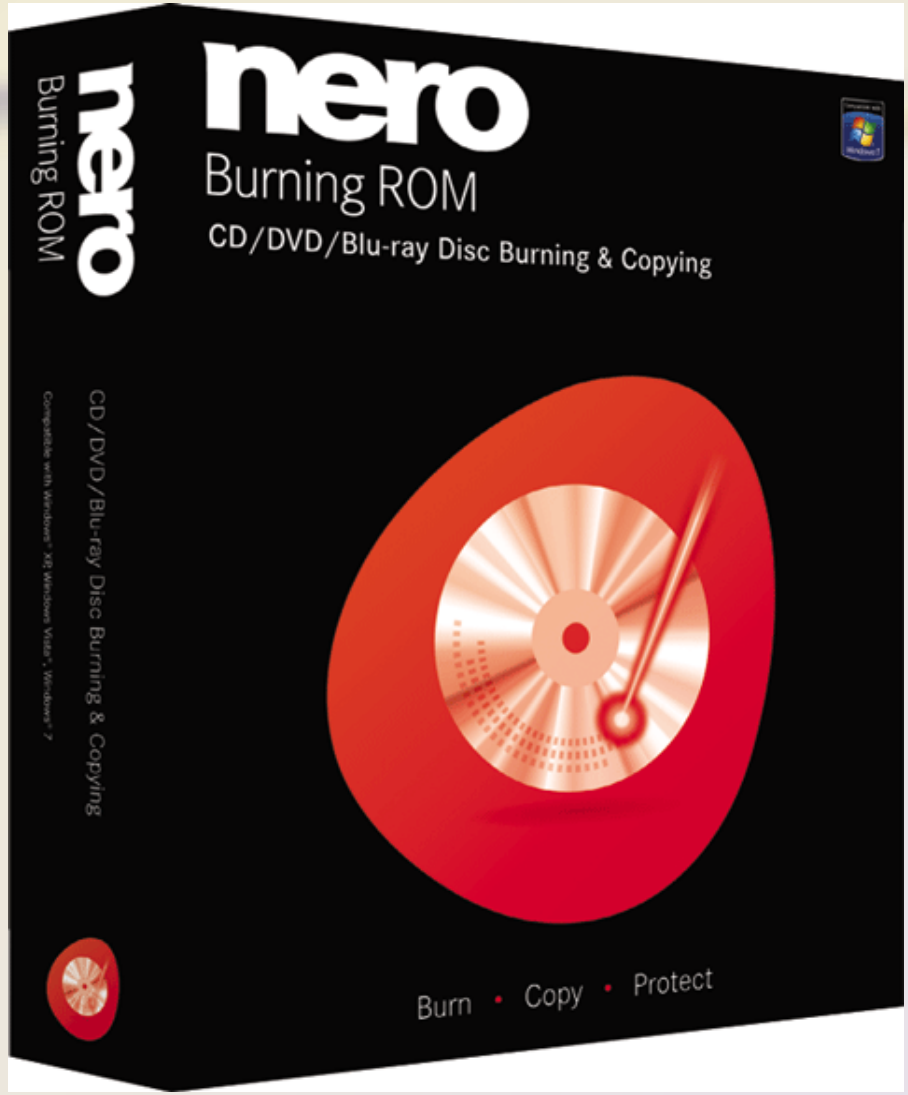
Utility Programs

- A **media player** allows you to view images and animation, listen to audio, and watch video files on your computer.

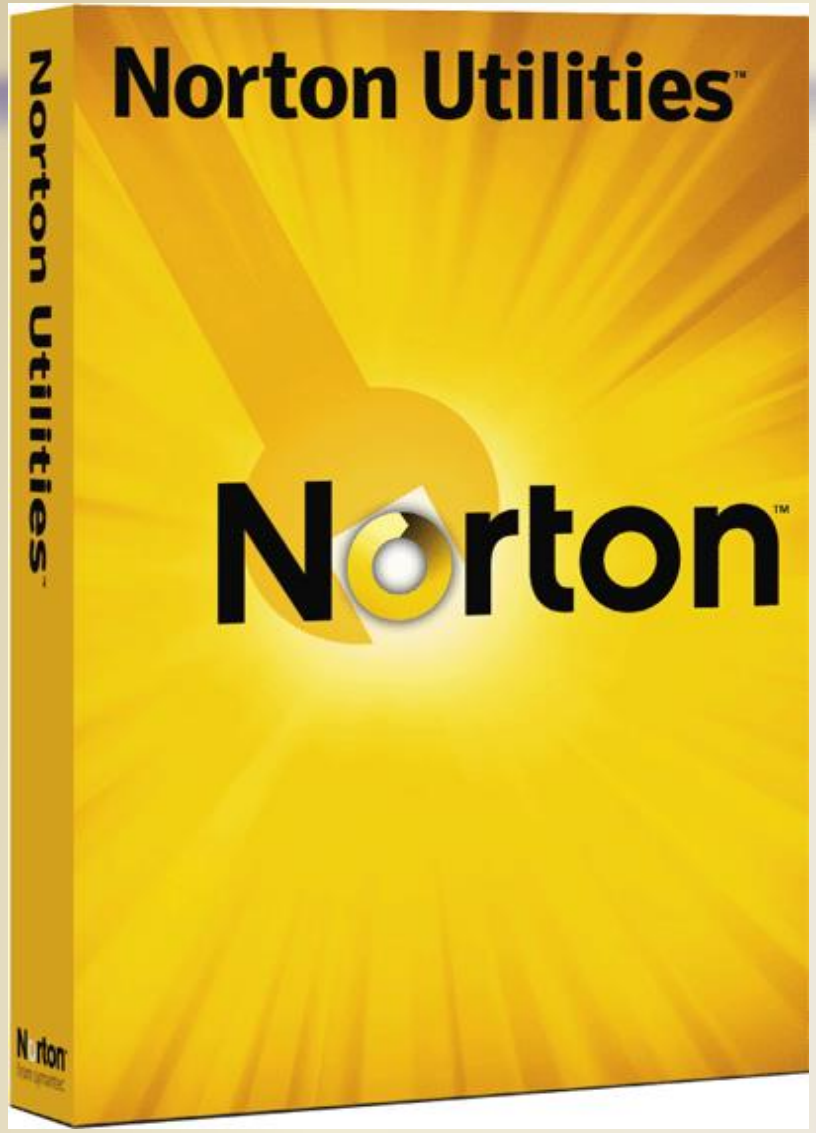


Utility Programs

Disc burning software writes text, graphics, audio, and video files on a recordable or rewritable optical disc.



Utility Programs



- A **personal computer maintenance utility** identifies and fixes operating system problems, detects and repairs disk problems, and includes the capability of improving a computer's performance.

Summary

What is a software and types of software

Functions common to most operating systems

Variety of stand-alone operating systems, server operating systems, and embedded operating systems

Several utility programs